Supplementary Statistical Analyses

Cross-Orientation Suppression in Macaque V1

Prepared for discussion with Wyeth

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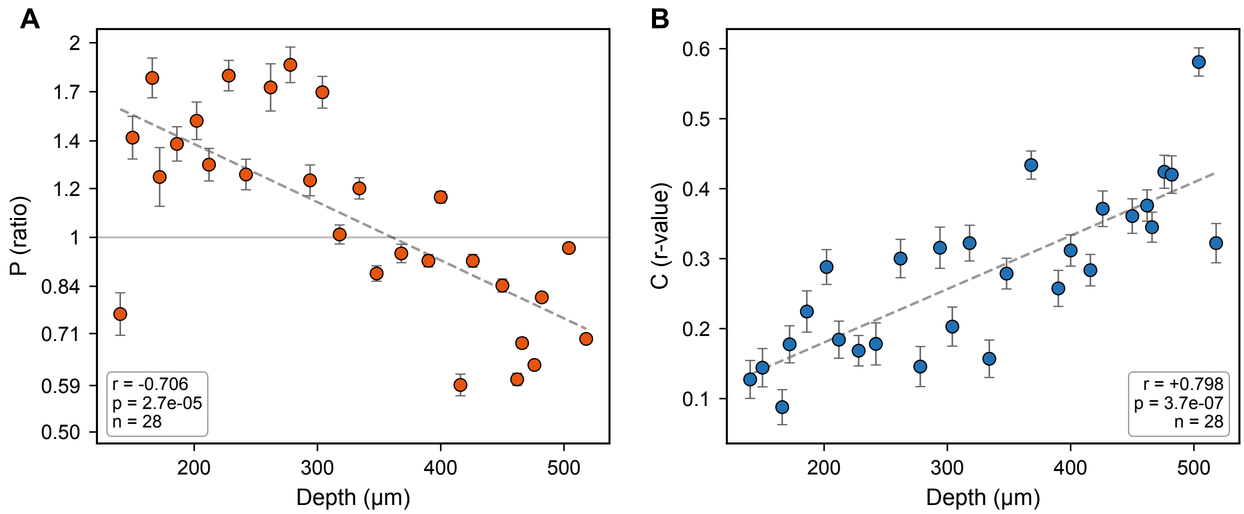
Overview

The main finding is that $\boldsymbol{P}$ (plaid suppression ratio) decreases with cortical depth—facilitation in superficial L2/3, suppression at depth. Four additional analyses defend this finding from every angle a reviewer could attack.

Analysis	**Reviewer question**	**Our answer**
Partial correlations	Is it a confound?	No—survives all controls
Mixed-effects model	Only 28 sites?	Holds at ROI level, $p = 8 \times 10^{-10}$
Mediation	Why does P change?	SF explains 40%, 60% unexplained
Subpopulation splits	Only one cell type?	Universal across all types

1 The Main Result

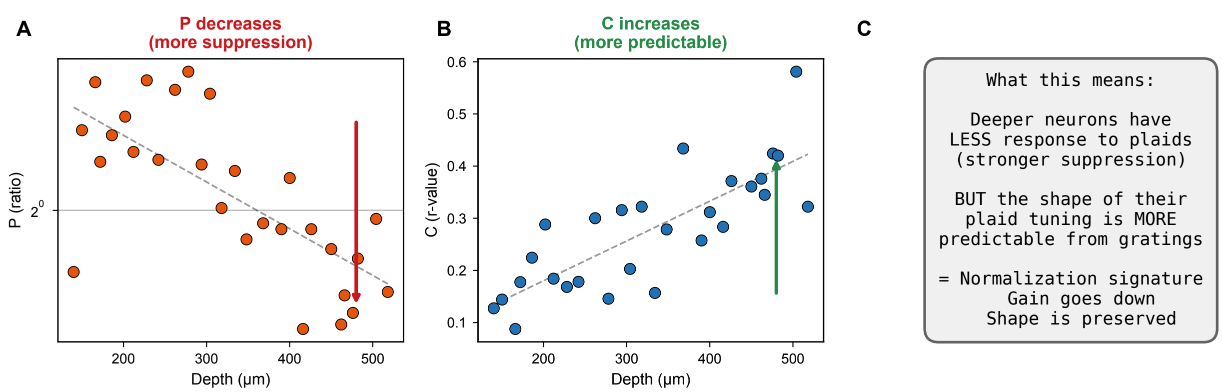


Panel A: ***P*** vs. depth ($r = -0.706$, $p = 2.7 \times 10^{-5}$, $n = 28$). Each point is one imaging site (mean across ∼170 ROIs). P is plotted on a log axis—$P > 1$ means facilitation, $P < 1$ means suppression, $P = 1$ is linearity.

Shallow sites show facilitation ($P > 1$). Deep sites show suppression ($P < 1$). Crossover at ∼350 μm.

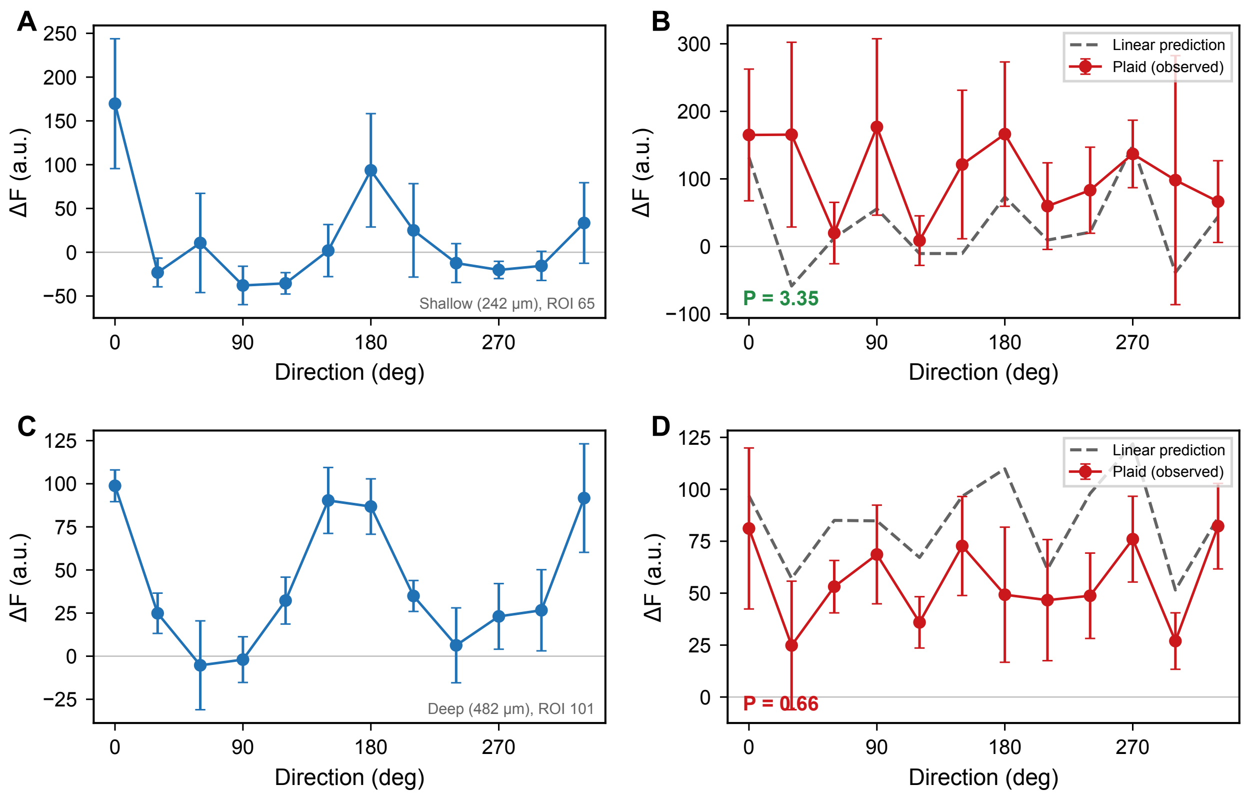
Panel B: ***C*** vs. depth ($r = +0.798$, $p = 3.7 \times 10^{-7}$). Even though deeper neurons are more suppressed, their plaid tuning *shape* is more predictable from the grating response. This is the normalization signature: gain decreases but shape is preserved.

The P–C dissociation (why this matters)



P goes down (A) while C goes up (B). This rules out a trivial explanation like "deep neurons just have noisier data." If it were noise, both P and C would degrade. Instead, the plaid shape becomes *more* predictable even as the amplitude drops. That's the hallmark of divisive normalization: it scales the gain but preserves the tuning curve shape.

2 Example Tuning Curves



Top row: Shallow site (242 μm). The observed plaid (red) exceeds the linear prediction (gray dashed). $P = 3.35$—strong facilitation.

Bottom row: Deep site (482 μm). The observed plaid falls below the prediction. $P = 0.66$—suppression.

This is what the main effect looks like in single neurons.

3 Partial Correlations

"Is the P-depth effect just a confound?"

We remove the influence of each confound from both P and depth, then re-test the correlation. If it survives, the confound doesn't explain the finding.

How it works

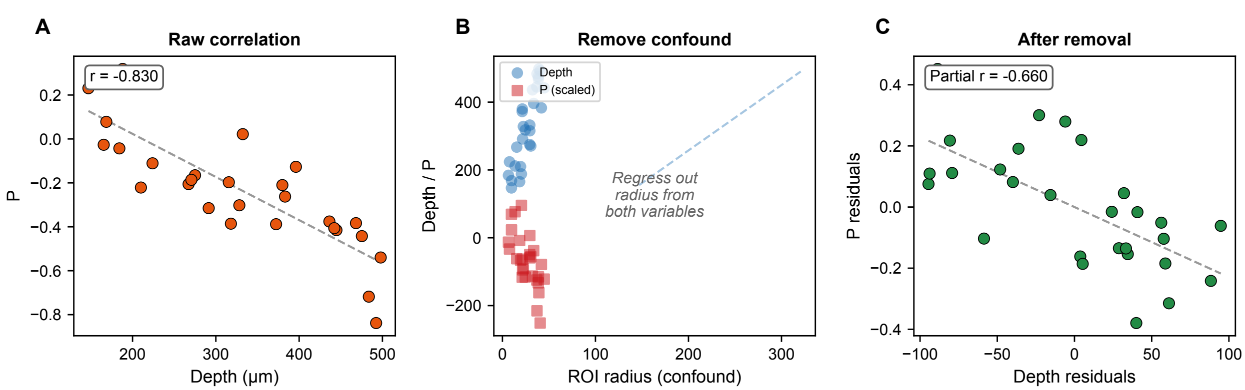
1. Regress depth on the confound $\rightarrow$ get depth residuals
2. Regress P on the confound $\rightarrow$ get P residuals
3. Correlate the residuals
4. If still significant $\rightarrow$ the confound doesn't explain the P-depth relationship

Results

Confounds removed	Partial r	p-value
None (raw correlation)	-0.706	2.7×10^{-5}
ROI radius	-0.751	< 0.001
All (radius + SNR + OSI + SF)	-0.520	0.005

The P-depth effect survives removing every confound we can measure. No single variable or combination explains it away.

Visual: how partial correlations work



A: Raw P vs depth—strong negative correlation. **B**: Both depth and P correlate with ROI radius (the confound). We regress radius out of both. **C**: After removal, the residuals still correlate. The confound didn't explain the effect.

4 Mixed-Effects Models

"You only have 28 data points."

The site-level correlation uses 28 means. A mixed-effects model uses all 4,785 ROIs while properly accounting for the nested structure.

The model

$$P_{ij} = \beta_0 + \beta_1 \cdot \text{Depth}_j + u_j + \varepsilon_{ij}$$

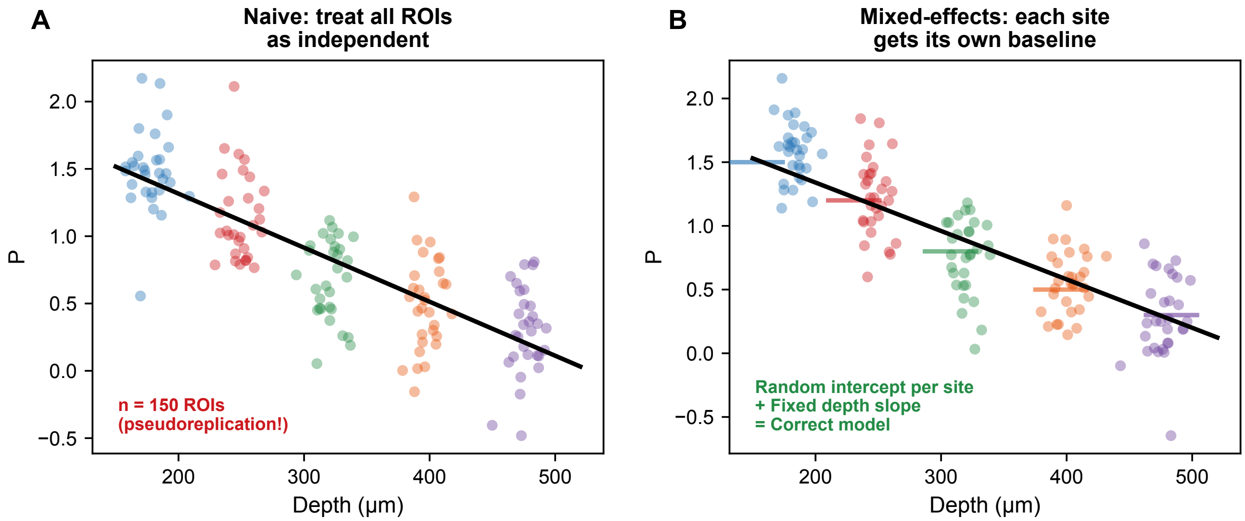
- i = ROI, j = site
- β_1 = fixed effect of depth (what we care about)
- u_j = random intercept per site—absorbs site-to-site differences unrelated to depth (expression level, focus quality, etc.)
- It's like a within-subjects ANOVA: each site is a "subject," depth is the treatment

Results

Model	$\boldsymbol{\beta}_{\textbf{depth}}$	p**-value**
$P \sim$ Depth + (1|Site)	-9.30	7.9×10^{-10}
$P \sim$ Depth + Radius + SNR + (1|Site)	-8.64	2.2×10^{-8}
$C \sim$ Depth + (1|Site)	$+0.086$	1.7×10^{-11}
$C \sim$ Depth + Radius + SNR + (1|Site)	$+0.036$	0.151 (NS)

P survives all controls ($p = 2.2 \times 10^{-8}$). C loses significance after controlling for radius and SNR—it's partially confounded by morphological factors. P is the stronger metric.

Visual: naive pooling vs. mixed-effects



A: Naive approach—treat every ROI as independent. This is pseudoreplication: 170 ROIs from the same site aren't 170 independent measurements of the depth effect. **B**: Mixed-effects—each site (color) gets its own baseline (horizontal bar = random intercept). The black line is the shared depth slope. This separates "site-to-site variability" from "the effect of depth," which is what we actually want to test.

5 Mediation Analysis

"OK the effect is real—but why? Is it just SF changing with depth?"

We control for one covariate at a time and measure how much the P-depth correlation drops. The bigger the drop, the more that variable *mediates* (explains) the relationship.

Results

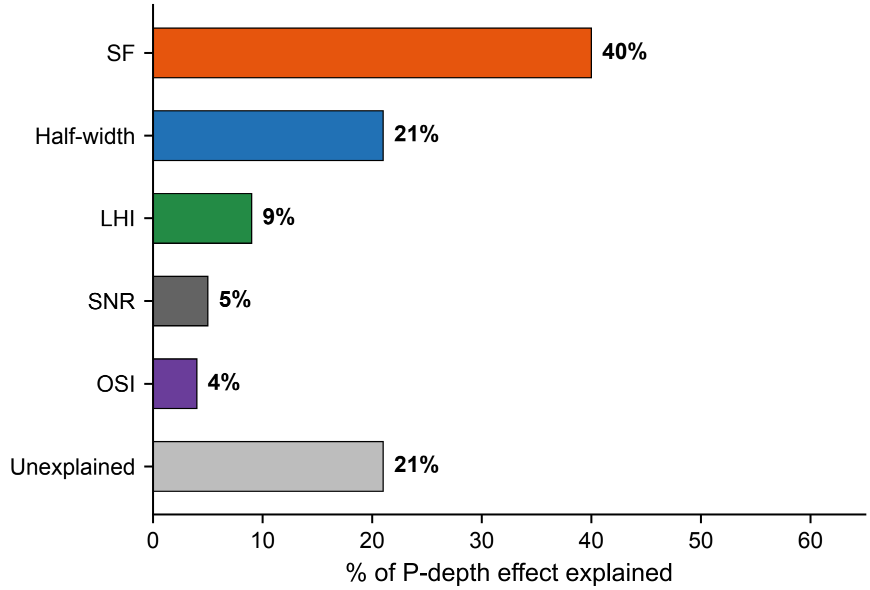
Control for	**Partial** r	**Drop from** -0.706	**% explained**
Nothing (baseline)	−0.706	—	—
Spatial frequency	−0.424	0.282	**40%**
Half-width (bandwidth)	−0.558	0.148	**21%**
LHI	−0.642	0.064	9%
SNR	−0.671	0.035	5%
OSI	−0.678	0.028	4%

The biological story

- **SF explains 40%**: deeper neurons prefer lower SF → larger receptive fields → broader normalization pools → stronger suppression
- **Bandwidth explains 21%**: broader tuning at depth → less selective normalization
- **60% is NOT explained by any measured covariate**—this is likely circuit architecture (inhibitory connectivity, cell-type composition) that changes with depth

This is what motivates the connectomics follow-up: the unexplained 60% is presumably in the wiring.

Visual: what each covariate explains



SF is the single biggest explainer at 40%. But the gray bar—the unexplained portion—is the most important. It means there's a genuine depth-dependent change in cross-orientation suppression that goes beyond what we can attribute to any measured functional property. The connectomics reconstruction will let us ask whether this remaining 60% is in the wiring.

6 Subpopulation Splits

"Maybe only one type of neuron drives this."

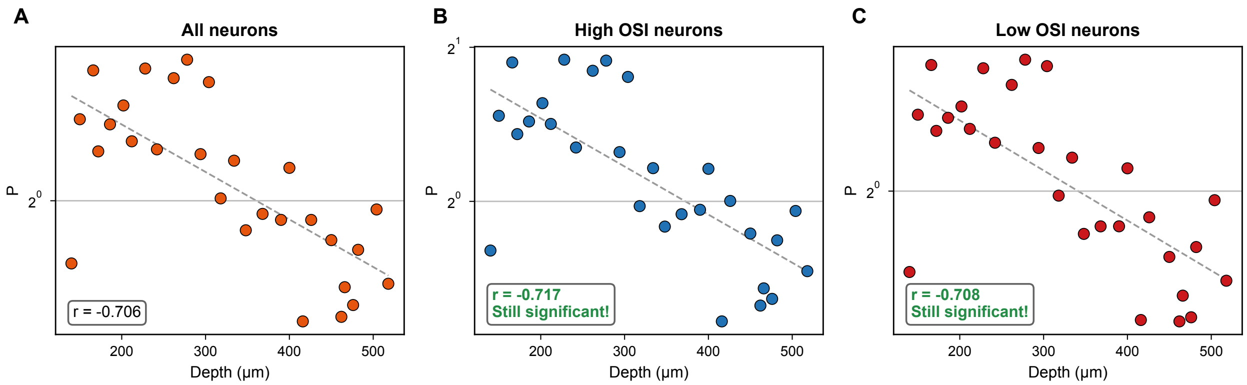
We split all ROIs by median OSI and median SF, then check the P-depth correlation in each half.

Results

Subpopulation	P-depth effect present?
High OSI neurons	✓
Low OSI neurons	✓
High SF preference	✓
Low SF preference	✓

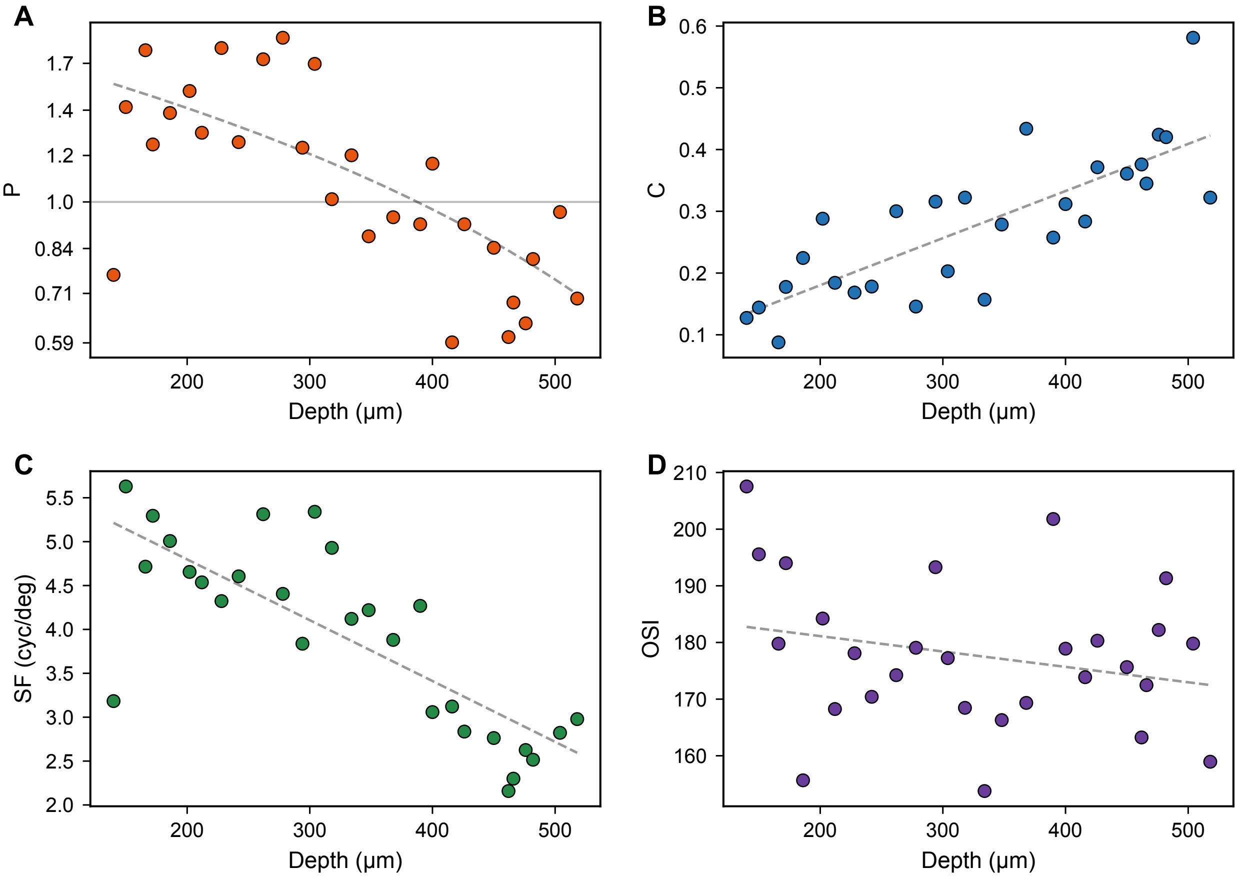
The effect is universal. Not driven by one neuron subclass.

Visual: the effect holds in every subgroup



A: All neurons together. **B**: High-OSI neurons only—still significant. **C**: Low-OSI neurons only—still significant. Same pattern for high/low SF splits. No matter how you slice the population, the depth gradient in P is present.

7 Depth Profile Summary



Eight metrics change coordinately with depth. This is not random—it's a systematic gradient in functional properties across L2/3.

Deeper in L2/3	Direction
ROI size	Increases
Baseline fluorescence	Increases
Orientation bandwidth	Broader
Preferred SF	Lower
OSI	Lower
LHI (local homogeneity)	Lower
P (suppression ratio)	More suppressive
C (shape similarity)	More predictable

Superficial L2/3: high SF, narrow tuning, high selectivity, homogeneous orientation maps, facilitation.

Deep L2/3: low SF, broad tuning, low selectivity, heterogeneous maps, suppression.

Where These Go in the Paper

Main Results text	Supplementary
P and C vs depth (Fig. 3)	Partial correlations table
Mixed-effects confirmation	Bootstrap CIs
Mediation: SF = 40%	Subpopulation splits
Depth profile summary	Morphology confounds
	ROI-level scatter

Bottom Line

The P-depth correlation is robust (partial correlations), confirmed at the single-neuron level (mixed-effects, $p = 8 \times 10^{-10}$), partially mechanistic (SF mediates 40%), universal (all subpopulations), and part of a coordinated depth gradient spanning 8 functional metrics. The unexplained 60% motivates the connectomics integration.